Sea-ice sub-cycling in the FESOM2 GPU port: what pays

N. Koldunov, 6 August 2026 — summary of the mEVP communication study

The one change that pays

Exchange the ice-velocity halo every 4th mEVP sub-cycle instead of after every one. mEVP runs 120 sub-cycles per model time step and currently exchanges after every one; this exchanges 30 times instead of 120 and lets the halo be up to three sub-cycles old. It is one conditional, switched on at run time, and it is an *approximation* — which is why most of the work went into measuring what it costs.

What it buys

Each mesh at a node count where the model still scales — not the largest available, which would flatter the result. Change of the whole model time step; GPU, same allocation, faster of two repetitions.

mesh	**operating point**	**standard** s/step	**every 4th** sub-cycle	**every 8th** sub-cycle
CORE2 (127 k nodes)	1 node (4 GPUs), knee at 2	0.0666	−12.6%	−14.4%
fArc (638 k)	4 nodes (16 GPUs), flat past 4	0.1173	−11.4%	−14.3%
DARS (3.2 M)	8 nodes (32 GPUs), still scaling	0.1720	−14.2%	−15.5%
NG5 (7.4 M)	16 nodes (64 GPUs), still scaling	0.2424	−8.5%	−9.9%

So: 11–14% off a production GPU run at $K = 4$ on three of the four meshes (8.5% on NG5), and it is not an artefact of running past the scaling limit — each row is that mesh's own sensible operating point. (Run past the limit the number is larger — −25.5% for CORE2 on 16 nodes — but nobody runs CORE2 on 16 nodes, and that number is not the one to quote.)

Caveat: this is a GPU result. On CPU the same change is worth −1 to −3.4%, because a CPU process holds far less of the mesh and the ice exchange is a much smaller share of the step.

What it costs

One-year CORE2 integrations on the CPU backend, which is exactly reproducible, so the whole difference is the approximation. Yardstick: two independent and both correct implementations of the same mEVP scheme — our C port and Fortran FESOM2 — over the same year.

pattern correlation, 1 year	SST / SSS / SSH	ice concentration	ice thickness	u_{ice}	v_{ice}
C port versus Fortran (yardstick)	1.00000	0.99999	0.99998	0.95438	0.93908
every 4th sub-cycle	**1.000000**	**0.999995**	**0.999994**	**0.963328**	**0.950003**
every 8th sub-cycle	1.000000	0.999992	0.999981	0.961359	0.944857

Every field stays inside the yardstick at $K \leq 8$. Ice velocity is the sensitive field and the margin there is only 1.1–1.5×, so smaller K is the defensible choice — the error is monotone in K, and $K = 4$ keeps about 87% of the speed of $K = 8$. Integrated quantities are untouched: Northern Hemisphere ice extent deviates by 0.23% of its seasonal amplitude at $K = 8$, and surface temperature, salinity and sea surface height correlate at 1.000000.

Not yet measured: accuracy at high resolution. The accuracy test is CORE2; the speed results are on fArc, DARS and NG5. $K = 4$ is recommended *scoped to CORE2* until a high-resolution year has been run.

What does not pay — measured, so we do not have to argue about it

Danilov's divergence-form reformulation (carry $\nabla\cdot\sigma$ at nodes instead of σ at elements). Exact — it reproduces $\nabla\cdot\sigma$ to 10^{-15} relative — but worth −0.3 to +0.1% on CPU and 0 to +1.6% on GPU. The sub-cycle is limited by communication and by the number of GPU operations issued, not by arithmetic or memory traffic, so removing element arrays removes work that was not costing anything: a control that deleted three element arrays outright changed the time by exactly 0.00%.

The exact wide halo — Danilov's double-halo proposal (exchange every K-th sub-cycle but recompute a K-deep ghost ring, so the answer is unchanged bit for bit). Certified exact, and worth −1.8 to −6.8% at the operating points above, against the approximation's −8.5 to −15.5%. It needs $K = 8$: at $K = 2$ it *costs* time, because the duplicated ring is not amortised. On CPU it never wins on any mesh — at 2048 processes on fArc the ring a process must advance is larger than its own sub-domain. It remains the right choice for anyone who needs the answer unchanged bit for bit.

Recommendation

Adopt the delayed exchange at $K = 4$ as an opt-in option for GPU production runs, after a high-resolution accuracy year. Keep the exact wide halo for anyone who needs the answer unchanged bit for bit. The divergence form is correct and elegant, but it is not a speed lever on this hardware.